

50 kWe Prototype CHP — Wood Fuel Demand for 24 h Operation

Prepared for the local forestry partner (meeting 2026-10-02). Estimate of the wood needed to run a first **50 kWe** biomass gasification CHP prototype **24 hours a day**. Three independent methods are compared, because no 50 kWe unit has been selected yet.

1. Fuel demand (dry-matter basis)

Method	Basis (source)	kg/h	t/day	t/yr at 8,000 h	Green chips/day (50% moisture)	Solid m ³ /day (314 kg/m ³)
A. Scaled from Tokuo's 10 kWe design	10 kWe needs 45.5 kWth input = 10.9 kg/h (meeting_tokuo_aomi.md)	54.5	1.31	436	2.6 t	4.2
B. Measured at Spa Hotel Abukuma	~1 kWh electric per kg pellet (Entrade units, running since 2018)	50	1.20	400	2.4 t	3.8
C. Forest Twin model	18.5 GJ/t, net electrical efficiency 0.13 (workforce plan)	74.8	1.80	599	3.6 t	5.7
<i>Reference: Mishima scenario</i>	~750 t/yr for ≤50 kWe (NIES simulation, not a built plant)	85	2.05	750	4.1 t	6.5

Planning figure: 1.3–1.8 t dry wood per day = 2.5–3.5 t fresh green chips per day = about 450–600 t/yr.

Method B is a measured plant (most reliable); Method C is the conservative upper bound.

2. How each figure was calculated

Method	Calculation
A	50 kWe ÷ 10 kWe × 10.9 kg/h = 54.5 kg/h; × 24 h = 1,308 kg/day; × 8,000 h = 436 t/yr
B	50 kWe ÷ 1 kWh/kg = 50 kg/h; × 24 h = 1,200 kg/day; × 8,000 h = 400 t/yr

Method	Calculation
C	$50 \text{ kWe} \div 0.13 = 384.6 \text{ kWth input}; \times 3.6 = 1,385 \text{ MJ/h}; \div 18.5 \text{ MJ/kg} = 74.8 \text{ kg/h}; \times 24 \text{ h} = 1.80 \text{ t/day}$
Green chips	$\text{dry mass} \div (1 - 0.50) = 2 \times \text{dry mass}$
Solid volume	$\text{dry mass} \div 314 \text{ kg/m}^3$ (sugi, Forest Twin density)

3. Drying heat (50% → 15% moisture)

- Water to evaporate $\approx 0.82 \text{ t}$ per tonne of dry wood.
- At 2.45 GJ per tonne of water (EIS 2024 paper), this is $\approx \mathbf{2.0 \text{ GJ per tonne of dry wood}}$.
- For Method A (1.31 t/day): $\approx 2.6 \text{ GJ/day} \approx \mathbf{30 \text{ kWth continuous}}$.
- The CHP should deliver $\approx 100 \text{ kWth}$ (about $2\times$ the electric output, per the Abukuma ratio), so drying would use **$\sim 30\%$ of the heat**, before real dryer losses.
- Applies only if the chosen unit needs dry chips (Volter $\leq 15\%$, Entrenco $\leq 12\%$). Wet-tolerant units (ESPE $\leq 45\%$, Esperia $\leq 40\%$) fold drying into the package.

4. Caveats

- Tokuo's sheet (21.8 t/yr) assumes only 2,000 h/yr of operation. **Do not quote it for 24 h operation.**
- Figures are gasifier input only. Chipping losses, actual wood species and actual moisture are not included.
- Method B is for pellets (dry); equivalent chip demand depends on chip moisture.
- Henry's Mie-plant experience may give better numbers.
- No unit or maker has been selected, so efficiency is an assumption in A and C.